Question:

John drives for 3 hours at a speed of 60 mph and then turns around because he realizes he forgot something very important at home. He tries to get home in 4 hours but spends the first 2 hours in standstill traffic. He spends the next half-hour driving at a speed of 30mph, before being able to drive the remaining time of the 4 hours going at 80 mph. How far is he from home at the end of those 4 hours?

Reasoning + Answer:

<think> To solve, first calculate the distance John initially travels away from home, then calculate the distance he covers on the return journey, and finally determine how far he is from home at the end. The initial distance is 3 hours * 60 mph. For the return journey, he spends 2 hours in traffic (no distance), 0.5 hours at 30 mph (distance = time * speed), and the remaining time at 80 mph (distance = time * speed). Add these to find the total distance he covers and then find the net distance from home. Initial distance from home = 3 * 60 = 180 miles. For the return, distance in traffic is 0. Standstill distance = 0.5 * 30 = 15 miles. Remaining time = 4 - 2 - 0.5 = 1.5 hours. Distance at 80 mph = 1.5 * 80 = 120 miles. Total distance towards home = 15 + 120 = 135 miles. Net distance from home = Initial distance - distance towards home = 180 - 135 = 45 miles. </think><answer>45</answer>